

JIAHUI XIN

+44 7757 243685 | jiahui.xin@mrc-bsu.cam.ac.uk | [jiahui-xin.github.io](https://github.com/jiahui-xin)

RESEARCH INTERESTS

Adaptive Clinical Trial Design; Covariate- and Response-Adaptive Randomization; Robust Inference; Precision Medicine; Markov Decision Process.

EDUCATION

- Renmin University of China, Institute of Statistics and Big Data** Sept. 2021 – June 2026 (Expected)
Ph.D. Candidate in Statistics Beijing, China
 - GPA: 3.84/4.0
 - Supervised by **Prof. Wei Ma**.
- University of Cambridge, MRC Biostatistics Unit** Jan. 2026 – Present
Visiting Ph.D. Student Cambridge, UK
 - Hosted by **Dr. David Robertson**.
 - Working on exact inference for covariate-adjusted response-adaptive (CARA) designs in clinical trials, in collaboration with Drs. David Robertson and Stef Baas.
- University of Science and Technology of China, Special Class for the Gifted Young** Sept. 2016 – June 2021
B.S. in Statistics Hefei, China
 - Major GPA: 3.95/4.3, Total GPA: 3.75/4.3

PAPERS

- On the Achievability of Efficiency Bounds for Covariate-Adjusted Response-Adaptive Randomization.**
Jiahui Xin and Wei Ma.
Statistical Methods in Medical Research, 2025.
- Cross-Validation Model Averaging under Covariate-Adaptive Randomization.**
Fuyi Tu, Jiahui Xin and Wei Ma.
Major revision at *Statistica Sinica*.
- Covariate-Adaptive Randomization with Many Strata: Asymptotic Theory and Statistical Inference.**
Jiahui Xin, Hanzhong Liu, Feifang Hu and Wei Ma.
Submitted to *Biometrika*; early version available on arXiv:2405.18856.
- Covariate-Adaptive Enrollment for Single-Arm Trials Using External Controls.**
Jiahui Xin, Hengjia Fang, Zhiyue Huang, Satoshi Yoshida and Wei Ma.
Working paper.

PRESENTATIONS

- Isaac Newton Institute for Mathematical Sciences, University of Cambridge** 2026
Poster presentation at Causal Inference in Biomedicine Workshop Bristol, UK
Title: Covariate-adaptive enrollment for single-arm trials using external controls
- Isaac Newton Institute for Mathematical Sciences, University of Cambridge** 2026
Lightning talk & poster presentation at Causal Inference in the Real World Workshop Cambridge, UK
Title: Covariate-adaptive enrollment for single-arm trials using external controls
- MRC Biostatistics Unit, University of Cambridge** 2026
Invited talk at Efficient Study Design Discussion Group Cambridge, UK
Title: Covariate-adaptive randomization with many strata: Asymptotic theory and statistical inference
- Drug Information Association** 2025
Invited talk at Quantitative Science Forum Nanjing, China
Title: Covariate-adaptive enrollment for single-arm survival trials using external controls
- Chinese Statistical Association of Young Statisticians** 2025
Travel award & oral presentation at Ph.D. Student Forum Jinan, China
Title: Inference under covariate-adaptive randomization with many strata

INTERNSHIP EXPERIENCE

- Astellas Pharma  (Clinical Biometrics Function)** June 2025 – Aug. 2025
Clinical Biostatistics Intern, Adaptive Strategy Beijing, China
 - Mentors: **Dr. Zhiyue Huang and Satoshi Yoshida**.
 - Developed adaptive enrollment strategies for single-arm survival trials under practical covariate shift.
 - Presented project results in internal seminars, including a China–Japan joint statistics meeting and a final internship presentation within the Clinical Biometrics Function.

TEACHING EXPERIENCE

- **Adaptive Design and Sequential Learning** Spring 2025
Teaching Assistant, Renmin University of China
- **Advanced Statistical Computation** Fall 2024
Teaching Assistant, Renmin University of China
- **Advanced Applied Statistics and Data Analysis** Spring 2024
Teaching Assistant, Renmin University of China
- **Statistical Models and Inference** Fall 2022
Teaching Assistant, Renmin University of China

OTHER EXPERIENCE

- **Causal Analyses of Complex Experiments** Dec. 2025
Yau Mathematical Sciences Center, Tsinghua University Short course by Prof. Peng Ding (UC Berkeley)
- **Causal Inference and A/B Testing** July 2024 – Aug. 2024
Guanghua School of Management, Peking University Business Analysis Workshop by Tencent
- **Topics and Recent Advances in Causal Inference** June, July & Dec. 2023
Yau Mathematical Sciences Center, Tsinghua University Short course by Prof. Peng Ding (UC Berkeley)
- **Semiparametric Learning** July 2023
Institute of Statistics and Big Data, Renmin University of China Short course by Prof. Yanyuan Ma (Penn State)

SKILLS

- **Programming & Tools:** R, C++, Python, Julia, Linux, Git

REFERENCES

- **Dr. Wei Ma** 
Associate Professor
Institute of Statistics and Big Data
Renmin University of China
59 Zhongguancun Street
Beijing 100872, China
Phone: +86 10 6251 2209
Email: mawei@ruc.edu.cn
- **Dr. Hanzhong Liu** 
Associate Professor
Department of Statistics and Data Science
Tsinghua University
30 Shuangqing Road
Beijing 100084, China
Phone: +86 10 6278 0575
Email: lh2016@tsinghua.edu.cn
- **Dr. Zhiyue Huang** 
Statistical Science Lead
Development Division Statistical & RWD Science
Astellas Pharma
No.8 Jianguomenwai Avenue
Beijing 100022, China
Phone: +86 186 1855 4065
Email: zhiyue.huang@astellas.com
- **Dr. Feifang Hu** 
Professor and Department Chair
Department of Statistics
George Washington University
801 22nd St. NW
Washington, DC 20 052, USA
Phone: +1 804 310 0383
Email: feifang@gwu.edu
- **Dr. David Robertson** 
Assistant Research Professor
MRC Biostatistics Unit
University of Cambridge
Robinson Way
Cambridge CB2 0SR, UK
Email: david.robertson@mrc-bsu.cam.ac.uk